

Regression Discontinuity Design

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We have the following local linear regression model used in Sharp RDD:

$$Y = \alpha_0 + \alpha_1 \cdot S + \alpha_2 \cdot (R - C) + \alpha_3 \cdot S \cdot (R - C) + \varepsilon$$

where:

- $S = 1$ if the unit is treated and $S = 0$ otherwise,
- R is the running variable,
- C is the cutoff.

We now interpret each parameter by plugging in specific values of S and $(R - C)$.

Interpreting α_0

Set $S = 0$ (untreated) and $R - C = 0$ (unit exactly at the cutoff):

$$\begin{aligned} Y &= \alpha_0 + \alpha_1(0) + \alpha_2(0) + \alpha_3(0)(0) \\ &= \alpha_0. \end{aligned}$$

Thus,

$$\mathbb{E}[Y(0, U) \mid R = C] = \alpha_0.$$

Interpretation: α_0 is the *expected untreated outcome at the cutoff*. It is the intercept of the untreated potential outcome function evaluated at $R = C$.

Interpreting α_1

Set $S = 1$ (treated) and $R - C = 0$:

$$\begin{aligned} Y &= \alpha_0 + \alpha_1(1) + \alpha_2(0) + \alpha_3(1)(0) \\ &= \alpha_0 + \alpha_1. \end{aligned}$$

Hence,

$$\mathbb{E}[Y(1, U) \mid R = C] = \alpha_0 + \alpha_1.$$

We can rewrite α_0 as:

$$\mathbb{E}[Y(1, U) \mid R = C] = \underbrace{\mathbb{E}[Y(0, U) \mid R = C]}_{\alpha_0} + \alpha_1.$$

Subtracting the untreated outcome:

$$\mathbb{E}[Y(1, U) \mid R = C] - \mathbb{E}[Y(0, U) \mid R = C] = \alpha_1.$$

Interpretation: α_1 is the *treatment effect at the cutoff*—the jump in the conditional expectation at $R = C$. Under the smoothness assumption, this equals the causal RDD estimand.

Interpreting α_2

Now set $S = 0$ (untreated), but allow the running variable to vary:

$$\begin{aligned} Y &= \alpha_0 + \alpha_1(0) + \alpha_2(R - C) + \alpha_3(0)(R - C) \\ &= \alpha_0 + \alpha_2(R - C). \end{aligned}$$

This is a linear function in $(R - C)$ with slope α_2 .

Interpretation: α_2 is the *slope of the untreated potential outcome function*

$$E[Y(0, U) \mid R = r]$$

in a neighborhood around the cutoff. It describes how the untreated outcome changes with the running variable inside the chosen bandwidth.

Interpreting α_3

To see the role of α_3 , set $S = 1$ and allow $R - C$ to vary:

$$\begin{aligned} Y &= \alpha_0 + \alpha_1(1) + \alpha_2(R - C) + \alpha_3(1)(R - C) \\ &= \alpha_0 + \alpha_1 + (\alpha_2 + \alpha_3)(R - C). \end{aligned}$$

Thus:

- untreated slope = α_2 ,
- treated slope = $\alpha_2 + \alpha_3$.

So:

$$\alpha_3 = (\text{slope of treated}) - (\text{slope of untreated}).$$

Interpretation: α_3 is the *difference in slopes* between treated and untreated potential outcome curves near the cutoff. It allows the treated and untreated groups to have different linear trends with respect to $(R - C)$.

Summary of Parameter Interpretations

- α_0 : untreated expected outcome at the cutoff.

$$\alpha_0 = \mathbb{E}[Y(0, U) \mid R = C]$$

- α_1 : treatment effect at the cutoff (the RDD estimand).

$$\alpha_1 = \mathbb{E}[Y(1, U) \mid R = C] - \mathbb{E}[Y(0, U) \mid R = C]$$

- α_2 : slope of the untreated potential outcome function near the cutoff.

$$\alpha_2 = \left. \frac{\partial}{\partial r} \mathbb{E}[Y(0, U) \mid R = r] \right|_{r \approx C}$$

- α_3 : difference in slopes; the treated slope is $\alpha_2 + \alpha_3$.

$$\alpha_3 = \left(\left. \frac{\partial}{\partial r} \mathbb{E}[Y(1, U) \mid R = r] \right|_{r \approx C} \right) - \left(\left. \frac{\partial}{\partial r} \mathbb{E}[Y(0, U) \mid R = r] \right|_{r \approx C} \right)$$